

Requirements Specification Document

Rwanda NBSAP Monitoring Dashboard 2025–2030

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Document Control

Field	Value
Document Title	Requirements Specification Document
Project	Rwanda NBSAP Monitoring Dashboard
Prepared by	NBSAP Dashboard Development Team
Approved by	REMA — Department of Biodiversity & Landscape Management
Status	Final

1. Introduction

1.1 Purpose

This document defines the functional and non-functional requirements for the Rwanda National Biodiversity Strategy and Action Plan (NBSAP) Monitoring Dashboard. It serves as the authoritative reference for system capabilities, acceptance criteria, and stakeholder expectations.

1.2 Scope

The NBSAP Monitoring Dashboard is a national digital platform that enables Rwanda to track progress against its 22 NBSAP 2025–2030 targets, aligned to the Kunming-Montreal Global Biodiversity Framework (KM-GBF). The system replaces manual spreadsheet-based reporting with an automated, real-time, web-based monitoring platform.

1.3 Intended Audience

- REMA senior management and technical staff
- Ministry of Environment (MoE) oversight personnel

- Government ICT auditors
- System administrators and maintenance teams
- Development partners and international reporting bodies

1.4 Regulatory Context

Framework	Relevance
Convention on Biological Diversity (CBD)	Rwanda is a signatory; the system supports National Report preparation
Kunming-Montreal GBF (2022)	23 global targets; Rwanda's 22 national targets are mapped to these
Rwanda NBSAP 2025–2030	The system's primary policy instrument
Rwanda ICT Policy	System aligns with digital government transformation objectives
Data Protection Law (2021)	User data handling complies with Rwanda's data protection requirements

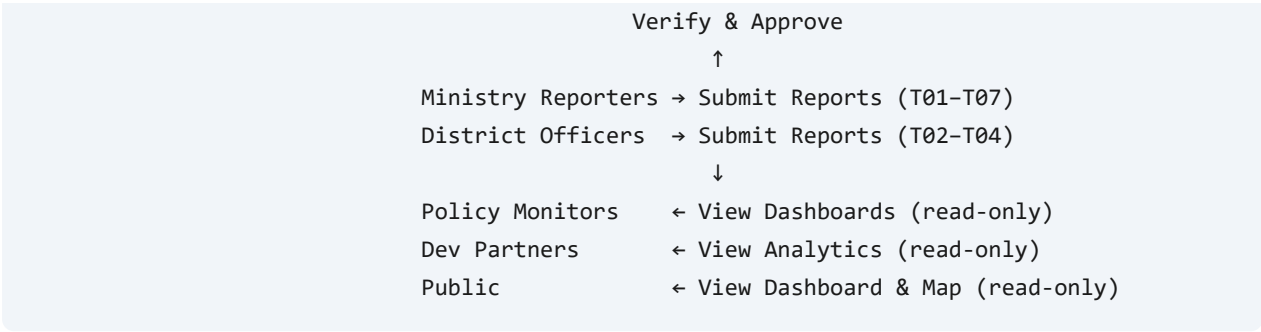
2. Stakeholder Analysis

2.1 Primary Stakeholders

Stakeholder	Role in System	Key Needs
REMA	System owner, administrator	Full oversight, data verification, user management, CBD reporting
Lead Government Ministries	Data providers (T01–T07 reports)	Structured reporting tools, verification feedback, analytics
District Reporting Officers	Field-level data entry	Simple forms, evidence upload, submission tracking
Policy Monitors	Strategic oversight	Read-only dashboards, national progress summaries
Development Partners	Programme alignment	Analytical views, indicator data access
General Public	Transparency audience	View-only access to dashboards and maps
CBD Secretariat	International reporting body	Structured indicator data in exportable formats

2.2 Stakeholder Interaction Model





3. Functional Requirements

3.1 User Authentication & Account Management

Ref	Requirement	Priority
FR-AUTH-01	The system shall authenticate users via email and password using JWT tokens	Must Have
FR-AUTH-02	Sessions shall auto-refresh without requiring re-login during active use	Must Have
FR-AUTH-03	Administrators shall be able to suspend accounts with reason and optional end date	Must Have
FR-AUTH-04	Suspended accounts shall auto-reactivate when the suspension end date passes	Should Have
FR-AUTH-05	Administrators shall be able to permanently deactivate accounts	Must Have
FR-AUTH-06	Users shall be able to request a role change, subject to administrator approval	Should Have

3.2 Role-Based Access Control

Ref	Requirement	Priority
FR-RBAC-01	The system shall enforce 6 distinct user roles with scoped permissions	Must Have
FR-RBAC-02	Each page and action shall check user role before granting access	Must Have
FR-RBAC-03	Only REMA Administrators shall access user management and audit logs	Must Have
FR-RBAC-04	Public Viewers shall have read-only access to dashboards, maps, targets, and indicators	Must Have

FR-RBAC-05	Role changes shall follow a formal request → review → approve/reject workflow	Should Have
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3.3 National Target Monitoring

Ref	Requirement	Priority
FR-TGT-01	The system shall display all 22 NBSAP 2025–2030 targets with progress tracking	Must Have
FR-TGT-02	Each target shall show baseline data, implementation milestones, and current progress	Must Have
FR-TGT-03	Targets shall be organised by 4 KM-GBF Goals (A, B, C, D)	Must Have
FR-TGT-04	Target progress shall be auto-calculated from weighted tool report data	Must Have
FR-TGT-05	A 3-colour progress system (green/amber/red) shall indicate status visually	Must Have

3.4 Indicator Framework

Ref	Requirement	Priority
FR-IND-01	The system shall track ~80 indicators across 4 tiers: Headline, Component, Binary, Complementary	Must Have
FR-IND-02	Each indicator shall display baseline, current value, midterm target, and 2030 target	Must Have
FR-IND-03	Indicators shall be linked to their parent NBSAP target	Must Have
FR-IND-04	Indicator status (on-track / at-risk / behind) shall be tracked and displayed	Must Have
FR-IND-05	Indicators shall follow KM-GBF 3-tier reporting structure for CBD compatibility	Must Have

3.5 Reporting Toolkit (T01–T07)

Ref	Requirement	Priority
FR-RPT-01	The system shall provide 7 structured reporting tools covering all NBSAP sectors	Must Have
FR-RPT-02	T01: National Institutional Reporting — institution, budget, compliance score, policy progress	Must Have

FR-RPT-03	T02: Ecosystem & Habitat Monitoring — district, forest ha, wetland ha, land cover change	Must Have
FR-RPT-04	T03: Protected Area Management — area name, coverage, management effectiveness, illegal cases	Must Have
FR-RPT-05	T04: Human-Wildlife Conflict — incident type, district, species, HWC cases, mitigation	Must Have
FR-RPT-06	T05: Finance & Resource Mobilisation — budget allocated, disbursed, finance source	Must Have
FR-RPT-07	T06: Private Sector EIA Compliance — company, EIA status, sector, restoration commitments	Must Have
FR-RPT-08	T07: Community Engagement — community name, district, participants, traditional knowledge	Must Have
FR-RPT-09	Each report shall support file/evidence attachments (documents, images)	Must Have
FR-RPT-10	Submitted reports shall default to 'pending' status awaiting verification	Must Have

3.6 Data Verification Queue

Ref	Requirement	Priority
FR-VRF-01	All submitted reports shall enter a mandatory verification queue	Must Have
FR-VRF-02	Authorised reviewers shall be able to approve or reject submissions with written rationale	Must Have
FR-VRF-03	Approved reports shall automatically update national metrics via database triggers	Must Have
FR-VRF-04	Rejected submissions shall notify the original reporter with the rejection reason	Must Have
FR-VRF-05	If an approved report is later deleted, all metric effects shall be automatically reversed	Must Have

3.7 Dashboard & Visualisation

Ref	Requirement	Priority
FR-DSH-01	The dashboard shall display an executive summary of national biodiversity status	Must Have
FR-DSH-02	13 automated system metrics shall be displayed with real-time values	Must Have

FR-DSH-03	An interactive map shall show all 30 districts across 5 provinces	Must Have
FR-DSH-04	Map layers shall include protected areas, river networks, lakes, and species occurrences	Must Have
FR-DSH-05	District-level compliance tracking shall be visible on the map	Should Have
FR-DSH-06	Charts shall visualise indicator trends and target progress	Must Have

3.8 Geographic Information System (GIS)

Ref	Requirement	Priority
FR-GIS-01	The system shall render an interactive map of Rwanda with district boundaries	Must Have
FR-GIS-02	GeoJSON layers shall display protected areas, rivers, and lakes	Must Have
FR-GIS-03	GBIF species occurrence data shall be overlaid on the map	Should Have
FR-GIS-04	Users shall be able to toggle map layers on/off	Must Have
FR-GIS-05	District detail panels shall display compliance and reporting status	Should Have

3.9 Data Export & CBD Reporting

Ref	Requirement	Priority
FR-EXP-01	The system shall export data in CSV format for spreadsheet analysis	Must Have
FR-EXP-02	The system shall export formatted PDF reports for distribution	Must Have
FR-EXP-03	The system shall export structured JSON for system interoperability	Should Have
FR-EXP-04	Export formats shall support direct use in CBD National Report preparation	Must Have

3.10 Audit & Compliance

Ref	Requirement	Priority
FR-AUD-01	The system shall maintain an immutable, append-only audit log of all system actions	Must Have
FR-AUD-02	The audit log shall record user identity, action type, timestamp, and affected data	Must Have
FR-AUD-03	A risk register shall track identified biodiversity risks with severity levels	Should Have

FR-AUD-04	Compliance records shall track adherence to regulatory frameworks	Should Have
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3.11 External System Integration

Ref	Requirement	Priority
FR-INT-01	The system shall integrate with GBIF API for species occurrence data (Rwanda)	Must Have
FR-INT-02	The system shall integrate with RBIS (Rwanda Biodiversity Information System)	Should Have
FR-INT-03	GBIF API calls shall be rate-limited to 1 request/second (client-side enforcement)	Must Have
FR-INT-04	A Supabase Edge Function shall proxy GBIF requests to avoid CORS issues	Should Have

3.12 Notifications

Ref	Requirement	Priority
FR-NTF-01	In-app notifications shall alert users of relevant events (report status changes, role approvals)	Must Have
FR-NTF-02	Real-time WebSocket push shall deliver notifications without page refresh	Should Have

4. Non-Functional Requirements

4.1 Performance

Ref	Requirement	Target
NFR-PERF-01	Initial page load time	< 3 seconds on 4G connection
NFR-PERF-02	API response time for standard queries	< 500ms
NFR-PERF-03	Dashboard metrics refresh interval	Real-time via WebSocket
NFR-PERF-04	Concurrent user capacity	500 users
NFR-PERF-05	Report storage capacity	10,000+ reports

4.2 Security

Ref	Requirement
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NFR-SEC-01	All traffic shall be encrypted via HTTPS/TLS
NFR-SEC-02	Row-Level Security (RLS) shall enforce data access at the database level
NFR-SEC-03	Security headers (CSP, X-Frame-Options, X-Content-Type-Options) shall be applied
NFR-SEC-04	JWT tokens shall be used for authentication with automatic refresh
NFR-SEC-05	Passwords shall be hashed using bcrypt (via Supabase Auth)
NFR-SEC-06	All data modifications shall be logged in the immutable audit trail

4.3 Availability & Reliability

Ref	Requirement	Target
NFR-AVL-01	System uptime	99.5% annually
NFR-AVL-02	Database backup frequency	Daily (automated)
NFR-AVL-03	Recovery Point Objective (RPO)	24 hours
NFR-AVL-04	Recovery Time Objective (RTO)	4 hours

4.4 Usability

Ref	Requirement
NFR-USB-01	The system shall be responsive across desktop, tablet, and mobile devices
NFR-USB-02	The interface shall use consistent colour coding (green/amber/red) for status
NFR-USB-03	All interactive elements shall have ARIA labels for screen reader accessibility
NFR-USB-04	The system shall meet WCAG 2.1 Level AA colour contrast requirements

4.5 Maintainability

Ref	Requirement
NFR-MNT-01	The codebase shall use TypeScript for type safety and maintainability
NFR-MNT-02	Database schema changes shall be managed through numbered migration files
NFR-MNT-03	The system shall support zero-downtime deployments via CI/CD
NFR-MNT-04	Comprehensive technical documentation shall be maintained

4.6 Scalability

Ref	Requirement
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NFR-SCL-01	The architecture shall support horizontal scaling through serverless deployment
NFR-SCL-02	Static assets shall be served via global CDN for geographic distribution
NFR-SCL-03	Database queries shall be optimised with 20+ indexes on frequently accessed columns

5. Data Requirements

5.1 Data Sources

Source	Type	Frequency	Description
Manual Entry	Structured forms	Quarterly / Ad hoc	T01–T07 reporting toolkit submissions
GBIF API	Automated feed	Real-time	Species occurrence records for Rwanda
RBIS API	Automated feed	Real-time	Rwanda Biodiversity Information System data streams
GeoJSON Files	Static reference	At deployment	District boundaries, protected areas, rivers, lakes

5.2 Data Retention

Data Category	Retention Period	Rationale
Audit logs	Indefinite	Regulatory compliance
Submitted reports	Duration of NBSAP period (2025–2030)	National reporting obligation
User profiles	Duration of account + 1 year	Data protection compliance
Indicator data	Indefinite	Historical trend analysis

5.3 Data Quality Controls

- Structured forms** — All data entry via controlled fields with validation rules
- Mandatory verification** — No data affects national metrics until approved
- Audit trail** — Every modification is logged with user identity and timestamp
- Automated calculations** — Database triggers ensure consistency (no manual metric computation)
- Reversal on deletion** — Deleting an approved report auto-reverses its metric impact

6. Constraints & Assumptions

6.1 Constraints

Constraint	Description
Budget	System designed for minimal operational cost (free/low-tier cloud services)
Connectivity	Users in districts may have limited bandwidth; system optimised for low-bandwidth
Browser	IE11 not supported; modern browsers (Chrome 90+, Firefox 88+, Safari 14+, Edge 90+) required
Language	Current version is English-only

6.2 Assumptions

Assumption	Rationale
District officers have internet access	Rwanda's national broadband coverage exceeds 95% of urban areas
REMA will designate system administrators	Ongoing user and data management requires institutional commitment
GBIF API will remain publicly accessible	GBIF is a multilateral research infrastructure with long-term funding
Supabase will maintain service availability	Supabase provides managed PostgreSQL with 99.9% uptime SLA

7. Acceptance Criteria Summary

Category	Criterion
Authentication	Users can register, log in, and maintain sessions without re-authentication
RBAC	Each of the 6 roles can access only their authorised pages and actions
Reporting	All 7 reporting tools (T01–T07) accept, validate, and store submissions
Verification	The approval/rejection workflow operates correctly with metric auto-update
Dashboard	The executive dashboard displays all 13 automated metrics accurately
Map	The interactive map renders all 30 districts with toggleable GeoJSON layers
Export	CSV, PDF, and JSON exports produce correct, complete data
Audit	The audit log records all significant system actions immutably
Performance	The system loads within 3 seconds and supports 500 concurrent users
Security	RLS, HTTPS, CSP headers, and JWT authentication are all operational

